

Assignment - 01

Artificial Intelligence

22BCON1068

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Q1. Explain the steps involved in converting the propositional logic statement into CNF with suitable example.

Ans. Steps to convert propositional logic to CNF.

(i) Eliminate Implications and Bi-Implications:

- Replace implications ($\rightarrow$) with their equivalent disjunctions: $A \rightarrow B \equiv \neg A \vee B$

- Replace bi-implications ($\leftrightarrow$) with their equivalent conjunctions of implications:

$$A \leftrightarrow B \equiv (A \rightarrow B) \wedge (B \rightarrow A)$$

(ii) Push Negations Inward (De Morgan's law)

- $\neg(A \wedge B) \equiv \neg A \vee \neg B$

- $\neg(A \vee B) \equiv \neg A \wedge \neg B$

(iii) Distribute $\vee$ over $\wedge$:

- $A \vee (B \wedge C) \equiv (A \vee B) \wedge (A \vee C)$

ex: Convert the following to CNF:

$$\neg(P \wedge Q) \rightarrow (R \vee \neg S)$$

(i) Eliminate Implication:

$$\neg(\neg(P \wedge Q)) \vee (R \vee \neg S)$$

(ii) Push Negations Inward:

$$(P \vee \neg Q) \vee (R \vee \neg S)$$

(iii) Distribute $\vee$ over $\wedge$:

(This step is not needed in this case as there are no conjunctions within disjunctions.)

$$\text{Final CNF: } (P \vee \neg Q) \vee (R \vee \neg S)$$

2. Write first order logic statements for the following.

- (i) if a perfect square is divisible by a prime p then it is also divisible by square of p .
- (ii) Every perfect square is divisible by some prime.
- (iii) Alice does not like Chemistry and History.
- (iv) if it is Saturday and warm and Sam is in the Park.

Ans

(i) if a perfect square is divisible by a prime p then it is also divide by the square of p .

$$\forall x \forall p ((\text{PerfectSquare}(x) \wedge \text{Prime}(p) \wedge \text{DivisibleBy}(x, p)) \rightarrow \text{DivisibleBy}(x, p^2))$$

(ii) Every perfect square is divisible by some prime.

$$\forall x (\text{PerfectSquare}(x) \rightarrow \exists p (\text{Prime}(p) \wedge \text{DivisibleBy}(x, p)))$$

(iii) Alice does not like Chemistry and History.

$$\neg \text{Likes}(\text{Alice}, \text{Chemistry}) \wedge \neg \text{Likes}(\text{Alice}, \text{History})$$

(iv) If it is Saturday and warm, then Sam is in the Park.

$$\forall d ((\text{Saturday}(d) \wedge \text{Warm}(d)) \rightarrow \text{InPark}(\text{Sam}, d))$$

This reads as: "For all days d , if it is Saturday and it is warm on day d , then Sam is in the Park on day d ."

2. Translate the following English sentences to propositional logic:

propositions: (R)aining, Liron is (S)ick, Liron is (H)ungry, Liron is (HA)ppy, Liron owns a (C)ar. Liron owns a (D)og (a)

- (a) it is raining if and only if Liron is sick.
 (b) If Liron is sick then it is raining, and vice versa.
 (c) It is raining is equivalent to Liron is sick.
 (d) Liron is hungry but happy.
 (e) Liron either owns a car or a dog.
 Ans (a) it is raining if and only if Liron is sick:

$$R \leftrightarrow S$$

This means "Raining" is true if and only if "Liron is sick".

- (b) If Liron is sick then it is raining, and vice versa.

$$R \leftrightarrow S$$

this expresses the same idea as (a) in a more detailed form, showing the mutual dependence between "Raining" and "Liron being sick".

- (c) It is raining is equivalent to Liron is sick:

$$R \leftrightarrow S$$

This is the same as (a) and (b), showing that "Raining" and "Liron being sick" are logically equivalent.

- (d) Liron is hungry but happy:

$$H \wedge HA$$

This means both "Liron is hungry" and "Liron is happy" must

be true at the same time.

(e) Liron either owns a cat or a dog.

C V D

this indicates that Liron owns at least one of the two pets: a cat or a dog.

4. Explain the syntax and semantics for first order Predicate Logics.

Ans Syntax of First-Order Predicate Logic:-

- Variables: Represent objects (e.g., x, y, z).
- Constants: Specific objects (e.g., a, b, c).
- Predicates: Represent properties or relationships (e.g., $P(x)$ mean " x is a person", $\text{love}(x, y)$ mean " x loves y ").
- Functions: Map objects to other objects (e.g., $f(x)$ could represent "father of x ").
- Quantifiers:
 - Universal $\forall$: "for all", e.g., $\forall x P(x)$.
 - Existential $\exists$: "There exists", e.g., $\exists x P(x)$.
- Logical connectives: AND ($\wedge$), OR ($\vee$), Not ($\neg$), implied ($\rightarrow$).
- Semantics of FOPL:-
 - Domain: The set of all objects under consideration.
 - Interpretation: Assigns meanings to constants, predicates, and functions in the domain.
 - Satisfaction: Determines whether a formula is true in a given interpretation (e.g., $P(x)$ is true if x is a person).

- Universal quantifier ($\forall x$):- True if the formula holds for all elements in the domain.
- Existential quantifier ($\exists x$):- True if the formula holds for at least one element in the domain.

In short, syntax defines the structure of valid expressions, while semantics gives meaning based on an interpretation and domain.

5. Convert the following statement in to first order predicate logics.

(i) Mary loves everyone.

$$\forall x (\text{loves}(\text{Mary}, x))$$

(ii) Every one walks or talks.

$$\forall x (\text{Walks}(x) \vee \text{Talks}(x))$$

(iii) Every student who walks talks.

$$\forall x ((\text{student}(x) \wedge \text{Walks}(x)) \rightarrow \text{Talks}(x))$$

(iv) Someone walks and talks.

$$\exists x (\text{Walks}(x) \wedge \text{Talks}(x))$$

(v) Someone walks and someone talks

$$\exists x \text{ Walks}(x) \wedge \exists y \text{ Talks}(y)$$

(vi) Everyone who walks is calm.

$$\forall x (\text{Walks}(x) \rightarrow \text{Calm}(x))$$

(vii) If anyone cheats, everyone suffers.

$$\exists x (\text{cheats}(x)) \rightarrow \forall y (\text{suffers}(y))$$